

# YASH RAWAL

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## PROFESSIONAL SUMMARY

AI/ML Engineer & Technical Lead specializing in production LLMs, Agentic AI architectures, and low-latency RAG systems. Proven expertise in fine-tuning open-source models, designing multi-agent MCP tools, and mentoring engineering teams.

## EXPERIENCE

### Bigscal Technologies Pvt. Ltd.

Surat, India

AI/ML Engineer (Technical Lead)

Oct. 2024 – Present

- Served as AI Lead driving 100% end-to-end ownership across multi-tenant microservices, agent runtimes, and models.
- Deployed enterprise agentic platform serving 1M+ active ERP users across 120+ cloud tenants on Azure and AWS.
- Architected multi-agent A2A routing with LangGraph & SAP Joule 2.0, unifying SAP SuccessFactors, Blend, and MCP APIs.
- Engineered dynamic UI widget rendering contracts and capability layers, boosting throughput 3× and saving \$120K+/year.
- Fine-tuned and served Qwen2.5-7B and E5-large-v2 on NVIDIA GPUs using PyTorch, Unsloth, and vLLM inference endpoints.
- Built 4-level hierarchical RAG in Python indexing 10,000+ technical documents via PGVector achieving 90%+ Recall@10.
- Conceived and built Prompt Lab evaluation control plane to benchmark LLM outputs and track hallucination metrics.
- Architected dual-door policy guardrails enforcing tenant rules, jailbreak defense, PII masking, and write claims.
- Mentored and upskilled 5+ interns and junior engineers in PyTorch fine-tuning, agentic workflows, and LLM evaluation.

### Bigscal Technologies Pvt. Ltd.

Surat, India

AI/ML Intern

Apr. 2024 – Sep. 2024

- Engineered clinical NLP summarization pipelines for 116K+ medical records using 4-bit quantized Llama 3 and FastAPI.
- Built real-time WebSocket audio streaming for Indic ASR using NVIDIA NeMo (RNN-T and CTC decoding) and audio DSP.
- Constructed multilingual speech synthesis pipelines with Coqui TTS, formatting custom LJSpeech audio datasets via DSP.

## PROJECTS

### vLLM High-Performance Inference Engine | Python, PyTorch, Pydantic, AsyncIO

Open Source

- Merged PR #55475 fixing XML tool parser drops on /v1/chat/completions and engineered torchcodec compatibility guards.
- Resolved unhandled torchcodec multimodal video decoder crashes by engineering resilient FFmpeg error boundaries.

### Job Architecture Generator (Qwen2.5-7B SFT) | Python, PyTorch, Unsloth, vLLM

LLM Training

- Fine-tuned Qwen2.5-7B on 10K+ domain examples using Unsloth and deployed low-latency OpenAI-compatible endpoints.
- Trained 4-bit QLoRA adapters with custom chat templates, cutting GPU memory overhead by 60% during vLLM serving.

### E5-large-v2 Job & Skills Embedding Fine-Tuning | Python, PyTorch, Transformers

Model Training

- Fine-tuned E5-large-v2 (335M params) with triplet loss, boosting domain semantic retrieval relevance by 18% over OpenAI.
- Constructed hard-negative mining pipeline over Lightcast labor datasets to separate closely aligned skill taxonomies.

### Edge Language Model for Edge Hardware | Python, PyTorch, CUDA, TinyStories

Research Project

- Pretrained sub-100M autoregressive Transformer from scratch on 462M tokens with custom Pre-LN architecture (1.43 val loss).
- Engineered FragmentStream tiled memory attention algorithm preventing OOMs during long-context generation.

### Modular Agent Harness & Conversational Copilot | Python, FastAPI, pgvector

Project

- Architected modular Python agent framework implementing formal ReAct loops with step budgets and dynamic tool routing.
- Engineered persistent multi-tenant session state and automated checkpoint recovery across asynchronous worker threads.

### Hierarchical Document Intelligence & RAG Platform | Python, pgvector, GROBID

Project

- Engineered 4-level hierarchical RAG ingestion for complex documents using GROBID parsing and PGVector similarity search.
- Developed paragraph-level signature deduplication and versioning pipeline supporting 1,000+ domain reference books.

## TECHNICAL SKILLS

**Languages & Backend:** Python, SQL, FastAPI, PostgreSQL, Oracle, REST APIs, WebSockets, AsyncIO, Linux, Git, Docker

**AI & ML Engineering:** LLMs, Agentic AI, MCP, Tool RAG, SFT/LoRA, vLLM Serving, RLHF, Prompt Tuning

**Frameworks & Libraries:** LangChain, LangGraph, PyTorch, Hugging Face Transformers, vLLM, SAP Joule 2.0, NVIDIA NeMo

**Cloud & DevOps:** Azure, AWS, Docker, Kubernetes, CI/CD (GitHub Actions), Redis, PGVector, PyTest, LLM-as-a-Judge

## EDUCATION & CERTIFICATIONS

### Parul University

Gujarat, India

Master of Computer Applications (Artificial Intelligence & Machine Learning)

2022 – 2024

**Relevant Coursework:** Deep Learning, Natural Language Processing, Distributed Systems, Cloud Architecture, Algorithms

**Certifications:** Microsoft Certified: Azure AI Fundamentals (AI-900) | Microsoft Certified: Azure Fundamentals (AZ-900)